

PMGM 7023 MANAGEMENT ANALYTICS

Management and Strategy, Faculty of Business and Economics

Office hours: By appointment, KKL 1317

Course Website: HKU Moodle

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COURSE DESCRIPTION

Managers often make arbitrary decisions based on intuition, chance, preference, convenience, or tradition. We need a scientific framework to guide us in a conscientious, explicit, and judicious use of the best available evidence to make informed managerial decisions.

In the lectures, we will cover the following topics: (1) Evidence-based Management, (2) Business Analytics Frameworks, (3) Data Preparation, (4) Descriptive & Diagnostic Analysis, (5) Predictive & Prescriptive Analysis. In the tutorials, you will have hands-on opportunities to learn how to perform management analytics with the programming language R. Please note that this course is not designed for technicians, but for future managers who wish to develop their data literacy. It does not require prior knowledge of programming and data analytics.

COURSE OBJECTIVES

1. Demonstrate an understanding of concepts, theories, and frameworks of management analytics
2. Appreciate the implications of evidence-based decision-making, with particular reference to MNEs.
3. Identify and analyze problems and provide innovative solutions for these problems.
4. Develop communication and leadership skills via various course assessments.

PROGRAMME LEARNING OUTCOMES

PLO1: Acquisition and internalization of knowledge of the programme discipline

PLO2: Application and integration of knowledge

PLO3: Inculcating professionalism and leadership

PLO4: Developing a global outlook

PLO5: Mastering communication skills

COURSE LEARNING OUTCOMES

CLO1: Acquisition and internalization of knowledge

CLO2: Apply the knowledge to case studies and exercises

CLO3: Executing analytics projects with both rigor and relevance

CLO4: Integrate knowledge from multiple areas for problem-solving

PLO1; PLO2

PLO1; PLO2

PLO2; PLO3; PLO4

PLO2; PLO5

COURSE TEACHING AND LEARNING ACTIVITIES

T&L1. Interactive Lecture	30 hours	25%
T&L2. Class Exercises	20 hours	17%
T&L3. Team Projects	20 hours	17%
T&L4. Final exam and self-study	50 hours	42%
	120 hours	100%

GRADES (tentative)

	Points	Alignment
Final Exam	200 points~40%	CLO1; CLO2; CLO3; CLO4
Team Analytic Project	150 points~30%	CLO1; CLO2; CLO4
R Markdown	75 points~15%	CLO2; CLO3
Padlet Participation	75 points~15%	
Total-1000 points	500 points~100%	

COURSE GRADE DESCRIPTORS (tentative)

Course Final Grade	Tentative Scores	Grade Descriptors
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A+, A, A-	100-90%	•Demonstrate a strong understanding of all relevant knowledge •Handling questions professionally •High participation in discussions •Present arguments that have an element of originality •Achieve a standard of excellent performance in the exams with very accurate computation and very good analytical and problem-solving skills
B+, B, B-	89-80%	•Excellent performance in assignments •Demonstrate a good understanding of all relevant knowledge •Handling questions in a logical way •Good participation in discussions •Present arguments that go beyond the lecture and textbook •Achieve a standard of good performance in the exams with accurate computation and good analytical and problem-solving skills
C+, C, C-	79-70%	•Good performance in assignments •Demonstrate a basic understanding of the concepts involved •Fairly address questions as set •Some participation in discussions •Present arguments in a well-structured manner •Meet a standard of acceptable performance in the exams with reasonably accurate computation and acceptable analytical and problem-solving skills
D+, D, D-	69-60%	•Acceptable performance in assignments •Demonstrate a minimum understanding of the concepts involved •Barely address questions as set •Minimal or no participation in discussions •Present arguments in a marginally acceptable manner •Meet a standard of marginally acceptable performance in the exams with some errors in computation and barely adequate analytical and problem-solving skills
F	<60%	•Marginally acceptable performance in assignments •Demonstrate a poor understanding of the concepts involved •Unable or unwilling to handle questions •Minimal or no participation in discussions •Present arguments poorly •Fail to meet a standard of passing the exams with major errors in computation and inadequate analytical and problem-solving skills •Poorly performance in assignments

FINAL EXAM (200 points)

The final exam will be an open-book exam in the format of multiple-choice questions. We will test both the conceptual and analytic content we learned in this class. It will be conducted during the lecture time and no makeup sessions will be allowed.

TEAM PROJECT (150 points)

At the beginning of the semester, you will have the option to form a cluster (typically with two to four students), and then we will try our best to accommodate your preferences in the random team assignment. Everybody on your team will receive the same score but your score might be subject to an individual adjustment at the end of the semester. **Individual adjustments are topped at - 20% of the project grade, and all adjustments will be completely evidence-based.** To minimize political behaviors, you cannot request any individual adjustments for yourself, either bonus points or no or less deduction. Individual adjustments aim to identify uneven contributions and reallocate the scores within your team. The best team should expect no individual adjustment. To aim that, you will be invited to take an online survey to let me know your experience with the team. This survey will be kept strictly confidential, and I will NOT discuss your responses with your teammates or any other parties without your approval. I might follow up with you if I would like to know further details about your responses. **Failure to complete the survey will result in a 10-point off from your individual grade.**

You will work in groups to complete a written report (no presentation, just a report) based on a decision-making scenario and several sets of data we gave you. You need to use R to complete the

group project and attach the R Markdowns as appendices. The goal of this project is to practice making managerial decisions based on data analyses. More details will be announced in class.

R MARKDOWN (75 points)

You will work on the software R to practice data analysis in the tutorials. We have class demonstrations to show you how to code and class exercises that invite you to code on your own. When you are done with your own coding, you can save your coding as an R markdown file and submit it to Moodle. We will grade your R Markdown into “Plus”, “Check”, and “No Record”. The deadline for each R Markdown is before the start of the next lecture. Late submission will be graded as “No Record”. You can drop the lowest one and it will not be counted into your final grade.

PADLET (75 points)

You will participate in various activities using a digital platform called Padlet. At the end of each session, we will aggregate all of your digital activities during the session and assign a grade of "Plus," "Check," or "No Record." If you are intellectually present for the lecture, you should expect to receive an “Plus”. You can drop the lowest one.

NOTES

1. Please be aware that any announcements will be posted in Moodle, which should automatically be sent to your university email address. Please be sure to check that on a regular basis. Please feel free to email me with questions or concerns about the course and/or assignments. I will endeavor to reply to you within 24 hours. If not, please send me a reminder because I will not ignore a student’s email.
2. Most importantly, I expect you to adhere to strong ethical principles in your academic work. At a minimum, I expect no cheating, plagiarism, or falsification of any work you turn in for this class. **I take academic dishonesty very seriously, and I will do whatever it takes to correct such behaviors.** In return, you will expect me to provide you with a classroom environment that is conducive to learning; be well organized and prepared for each class lecture; provide an environment that is respectful of others’ opinions; provide reasonable guidance on all assessments; and treat you fairly and courteously.

TENTATIVE COURSE SCHEDULE (subject to change)

<u>Session</u>	<u>Date</u>	<u>Topics, assignments, & exams</u>
1	Sep 3	Course Intro; Evidence-based Management; Getting Started with R
2	Sep 6	Basics of Data; Data Preparation
3	Sep 10	Common Data Traps; Mean and Standard Deviation
4	Sep 13	Asking the Right Questions; Correlation and Data Visualization
5	Sep 20	Experimental Analytics; Mean Comparison
6	Sep 24	Survey Analytics; Measurement Theory
7	Sep 27	Business Analytic Framework; Regression
8	Oct 4	Forecasting Model; Regression-based Forecasting; Decision Tree
9	Oct 8	Team Project Due; Team Project Debriefing; Final Exam Review
10	Oct 15	In-class Final Exam